Evidence index (snapshot 2026-11-13, about 4:20 PM, final compile)

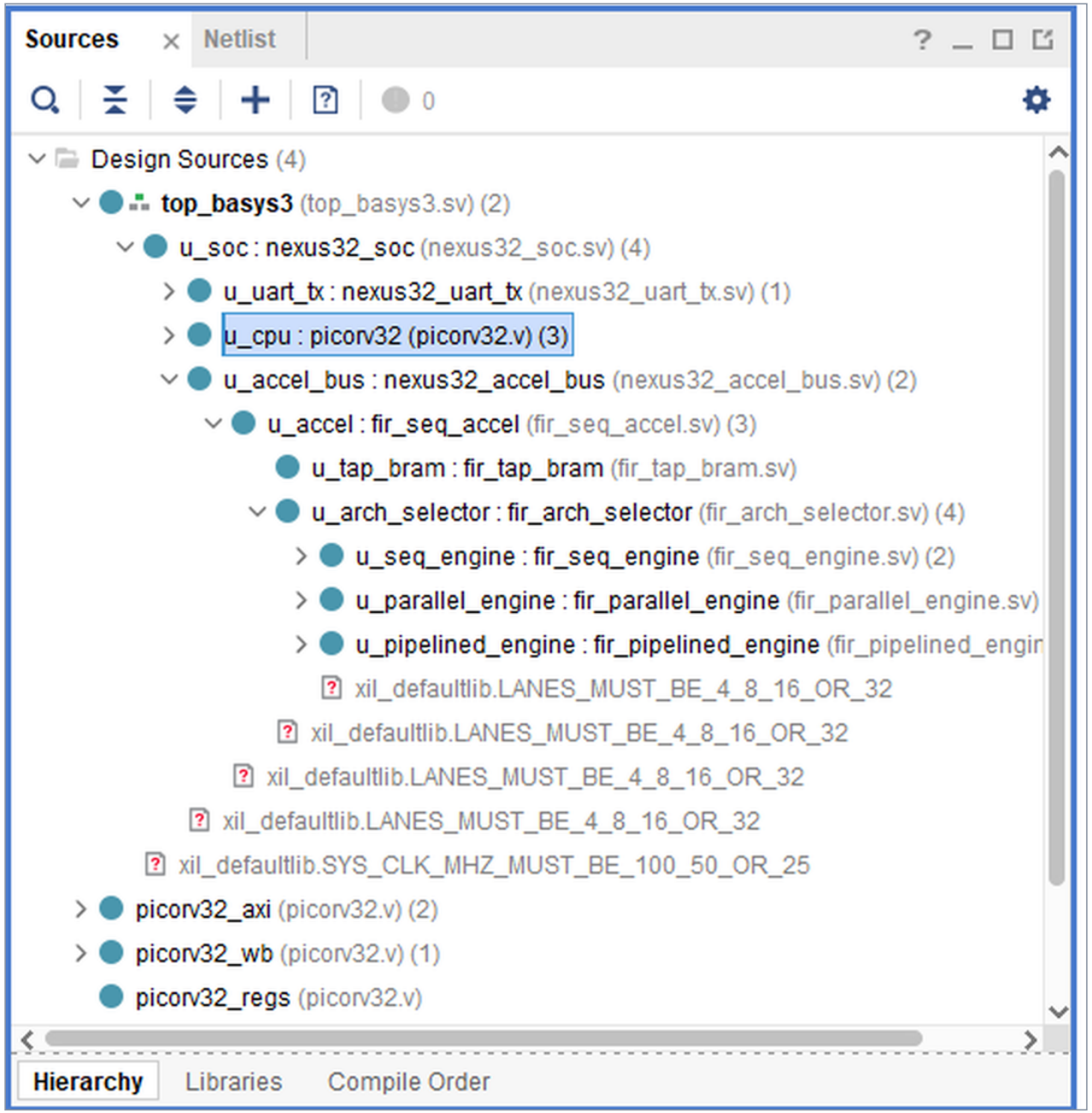
#	Screenshot	State
1	Sources hierarchy	**In this PDF**
2	Top-level RTL schematic	**In this PDF**
3	RTL schematic inside u_soc	**In this PDF**
4	Synthesized utilization	**In this PDF**
5	Routed timing summary	**In this PDF**
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7	Implemented utilization, Summary	**In this PDF**
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DRC	DRC report (0 errors)	**In this PDF**
Meth	Methodology report (0 critical)	**In this PDF**
9	Power estimate	**In this PDF**
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10	Device view, per-engine colours	**In this PDF**
10b	Colour legend	**In this PDF**
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S2	Simulation waveform	**In this PDF**
Board	LED photo and board photos	**In this PDF**
S1	Simulation console (3 x check=PASS)	**Not captured; optional, covered by board UART log (82 x PASS)**

Headline results (LANES=8, 50 MHz, xc7a35tcpg236-1)

Simulation PASS (cycles SEQ 14352 / PAR 5136 / PIPE 9232, checksum 0xb2be7f16). Synthesis, implementation and bitstream complete. Routed timing met: WNS +0.468 ns, WHS +0.014 ns, 0 failing endpoints. Implemented: 4695 LUT / 3614 FF / 68 DSP / 13 BRAM. Power estimate 0.194 W (MMCM 87% of dynamic); 163 DRC and 82 methodology warnings, no errors or critical items. On the Basys 3: LD0 and LD2 on, LD3 off, UART report NEXUS32 RESULT: PASS with cycles 14352 / 5136 / 9232 (identical to simulation), measured die temperature 33.3 C. Vivado power and junction temperature are estimates; the System Monitor reading is a measurement.

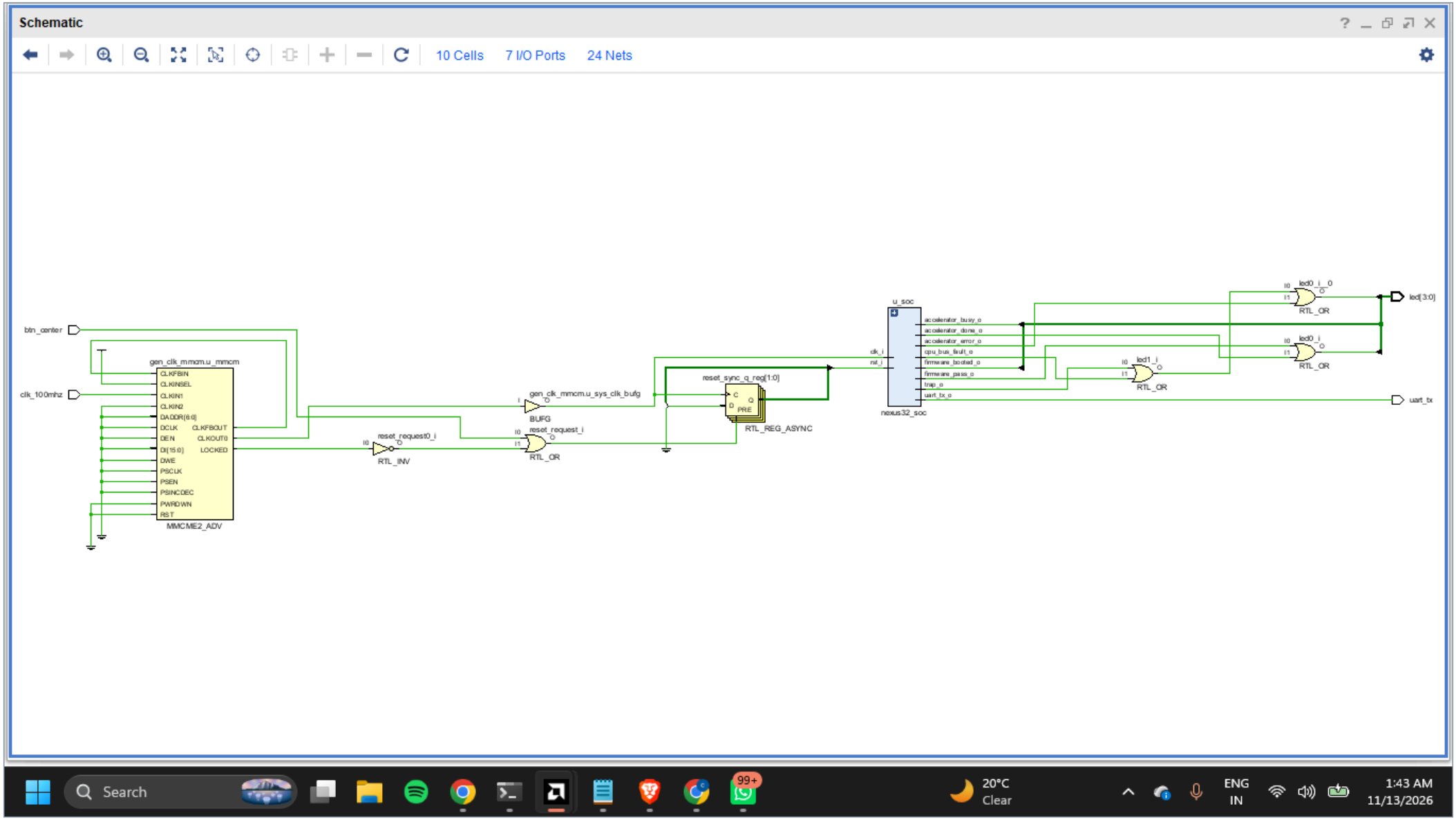
Screenshot 1: Sources hierarchy

Hierarchy view: top_basys3 > u_soc (nexus32_soc) > u_uart_tx, u_cpu (picorv32), u_accel_bus > u_accel (fir_seq_accel) > u_tap_bram, u_arch_selector > u_seq_engine, u_parallel_engine, u_pipelined_engine. The xil_defaultlib.*_MUST_BE_* entries are parameter-guard modules for LANES and SYS_CLK_MHZ (valid at LANES=8, 50 MHz). Cropped from the Vivado window.



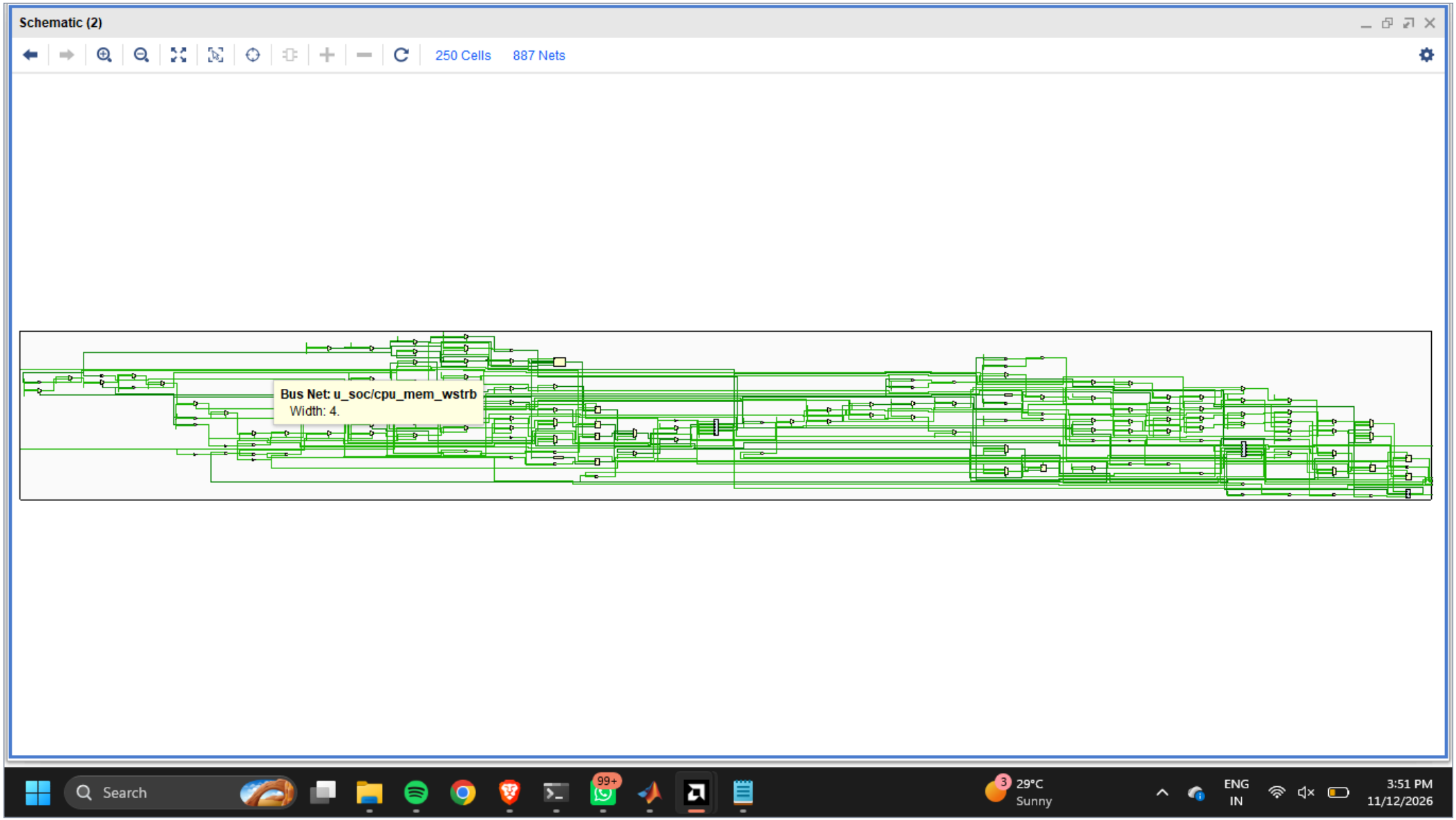
Screenshot 2: Top-level RTL schematic (elaborated design)

10 cells, 7 I/O ports, 24 nets. Inputs clk_100mhz and btn_center; outputs led[3:0] and uart_tx. Shows the MMCM (gen_clk_mmcm.u_mmcm), BUFG, reset synchroniser and the single u_soc (nexus32_soc) block.



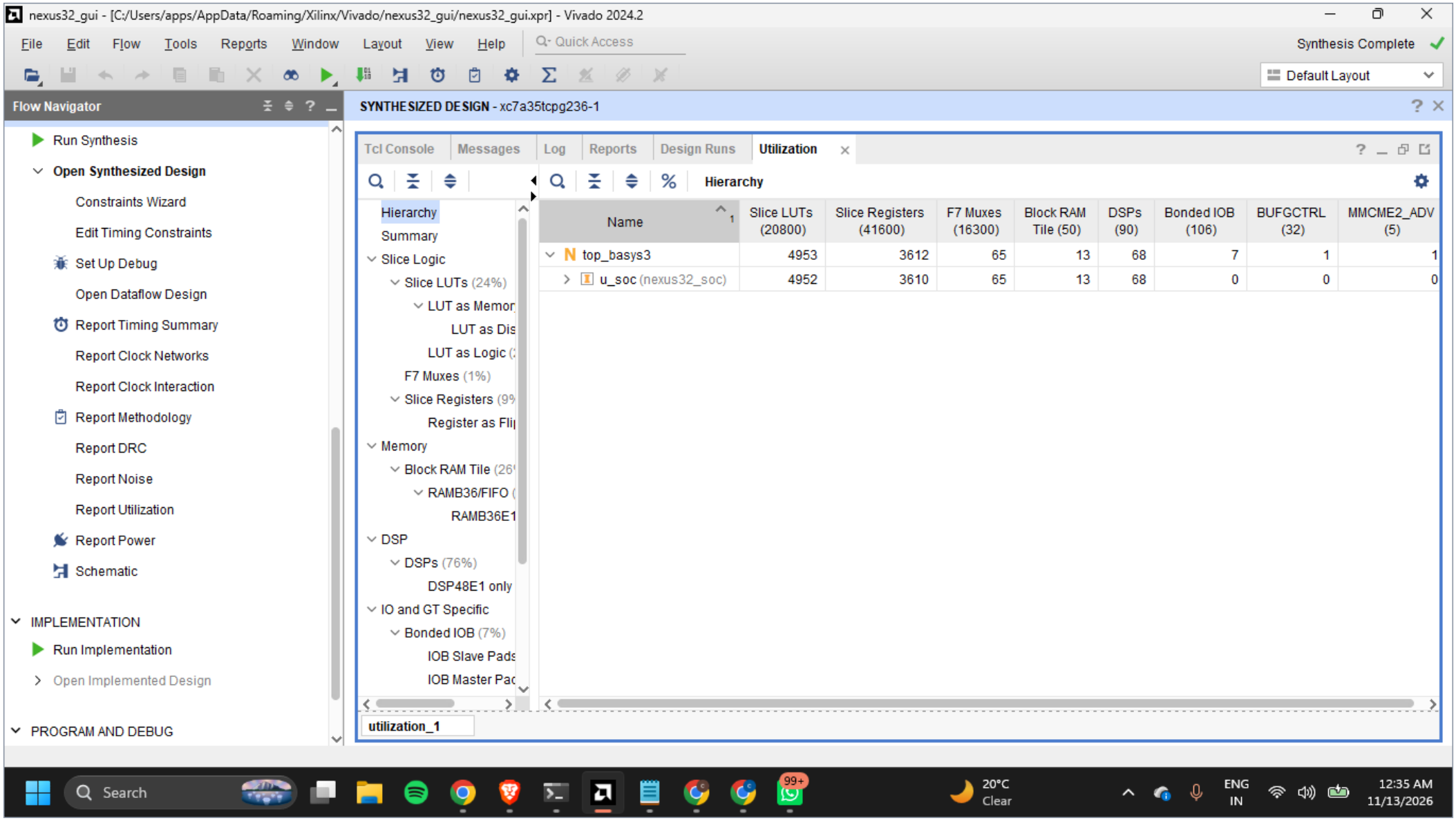
Screenshot 3: RTL schematic, inside u_soc

Vivado 2024.2, elaborated design: 250 cells, 887 nets. Full-size export in schematic_u_soc.pdf.



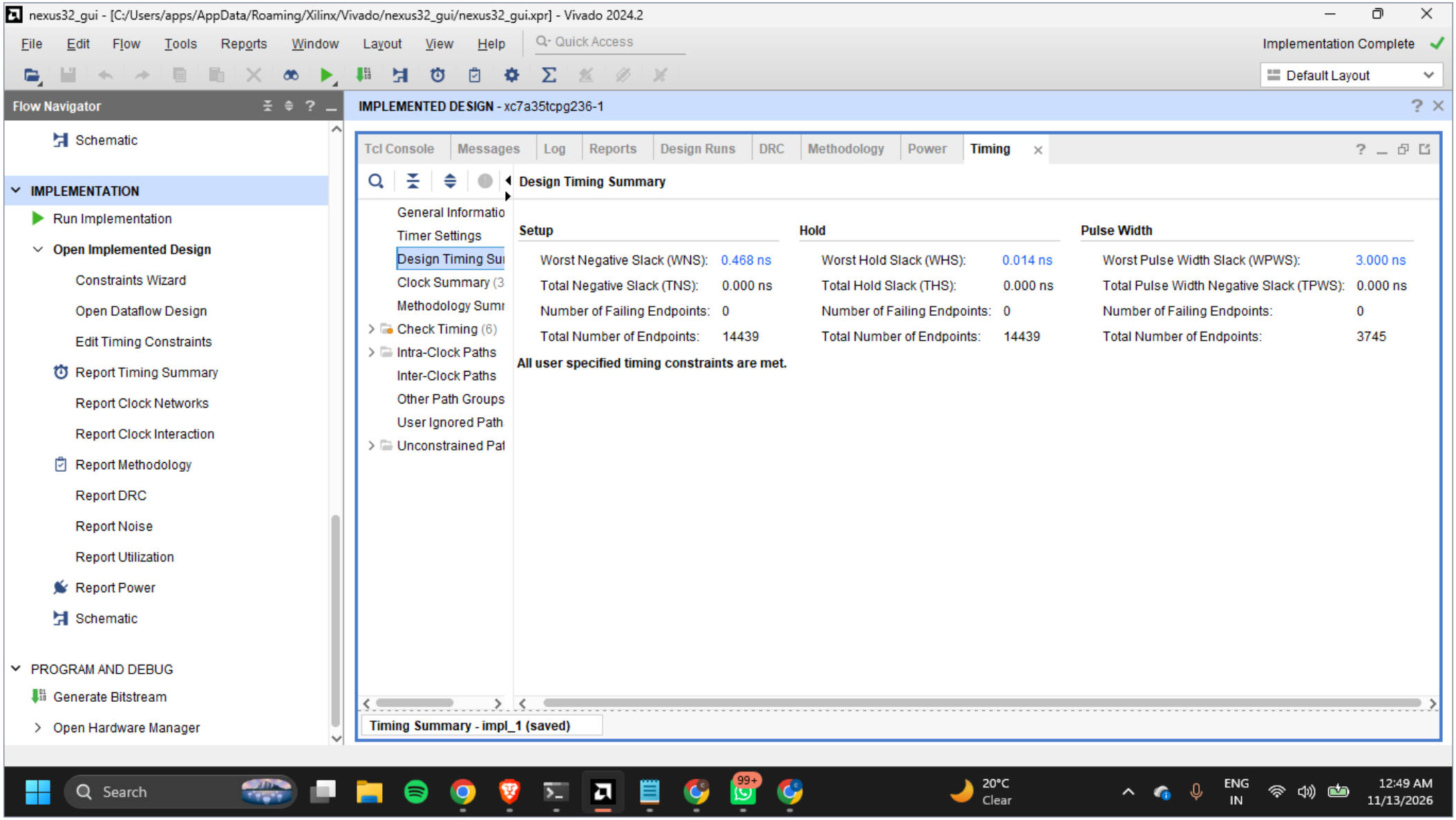
Screenshot 4: Synthesized utilization (LANES=8, 50 MHz)

LUT 4953 / 20800, FF 3612 / 41600, DSP 68 / 90, BRAM tile 13 / 50, IOB 7, BUFG 1, MMCM 1. Matches scripted run 1 (4955 / 3613 / 68 / 13). Hierarchy view; Summary view not captured.



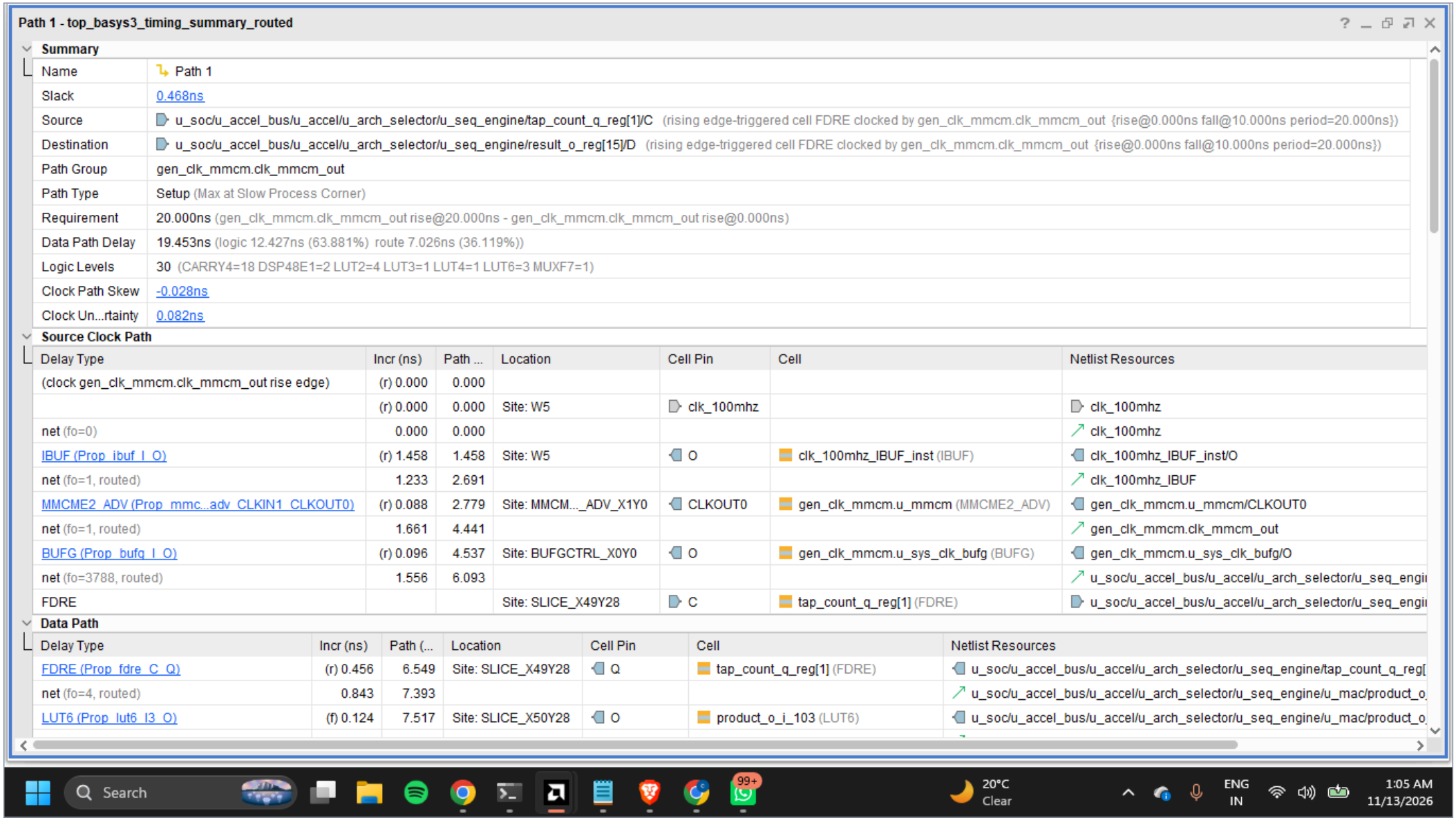
Screenshot 5: Routed Design Timing Summary (50 MHz)

WNS +0.468 ns, TNS 0, WHS +0.014 ns, THS 0, WPWS 3.000 ns. 0 failing endpoints. "All user specified timing constraints are met." Report: Timing Summary - impl_1 (saved), IMPLEMENTED DESIGN. Use WHS 0.014 ns; the scripted "WHS 3.000" is the pulse-width slack (script bug).



Screenshot 6: Critical path (routed, Path 1)

Slack +0.468 ns. Source u_seq_engine/tap_count_q_reg[1]/C, destination u_seq_engine/result_o_reg[15]/D. Requirement 20.000 ns. Data path delay 19.453 ns (logic 12.427 ns / route 7.026 ns). 30 logic levels: CARRY4 18, DSP48E1 2, LUT2 4, LUT3 1, LUT4 1, LUT6 3, MUXF7 1. Clock skew -0.028 ns, uncertainty 0.082 ns. The sequential engine does multiply, accumulate, shift and saturate in one cycle, which limits the SoC to 50 MHz.



Path 1 - top_basys3_timing_summary_routed

Summary

Name	Path 1
Slack	0.468ns
Source	u_soc/u_accel_bus/u_accel/u_arch_selector/u_seq_engine/tap_count_q_reg[1]/C (rising edge-triggered cell FDRE clocked by gen_clk_mmcm.clk_mmcm_out {rise@0.000ns fall@10.000ns period=20.000ns})
Destination	u_soc/u_accel_bus/u_accel/u_arch_selector/u_seq_engine/result_o_reg[15]/D (rising edge-triggered cell FDRE clocked by gen_clk_mmcm.clk_mmcm_out {rise@0.000ns fall@10.000ns period=20.000ns})
Path Group	gen_clk_mmcm.clk_mmcm_out
Path Type	Setup (Max at Slow Process Corner)
Requirement	20.000ns (gen_clk_mmcm.clk_mmcm_out rise@20.000ns - gen_clk_mmcm.clk_mmcm_out rise@0.000ns)
Data Path Delay	19.453ns (logic 12.427ns (63.881%) route 7.026ns (36.119%))
Logic Levels	30 (CARRY4=18 DSP48E1=2 LUT2=4 LUT3=1 LUT4=1 LUT6=3 MUXF7=1)
Clock Path Skew	-0.028ns
Clock Un...rtainty	0.082ns

Source Clock Path

Delay Type	Incr (ns)	Path ...	Location	Cell Pin	Cell	Netlist Resources
(clock gen_clk_mmcm.clk_mmcm_out rise edge)	(r) 0.000	0.000				
	(r) 0.000	0.000	Site: W5	clk_100mhz		clk_100mhz
net (fo=0)	0.000	0.000				clk_100mhz
IBUF (Prop_ibuf_I_O)	(r) 1.458	1.458	Site: W5	O	clk_100mhz_IBUF_inst (IBUF)	clk_100mhz_IBUF_inst/O
net (fo=1, routed)	1.233	2.691				clk_100mhz_IBUF
MMCME2_ADV (Prop_mmc...adv_CLKIN1_CLKOUT0)	(r) 0.088	2.779	Site: MMCM..._ADV_X1Y0	CLKOUT0	gen_clk_mmcm.u_mmcm (MMCME2_ADV)	gen_clk_mmcm.u_mmcm/CLKOUT0
net (fo=1, routed)	1.661	4.441				gen_clk_mmcm.clk_mmcm_out
BUFG (Prop_bufg_I_O)	(r) 0.096	4.537	Site: BUFGCTRL_X0Y0	O	gen_clk_mmcm.u_sys_clk_bufg (BUFG)	gen_clk_mmcm.u_sys_clk_bufg/O
net (fo=3788, routed)	1.556	6.093				u_soc/u_accel_bus/u_accel/u_arch_selector/u_seq_engi
FDRE			Site: SLICE_X49Y28	C	tap_count_q_reg[1] (FDRE)	u_soc/u_accel_bus/u_accel/u_arch_selector/u_seq_engi

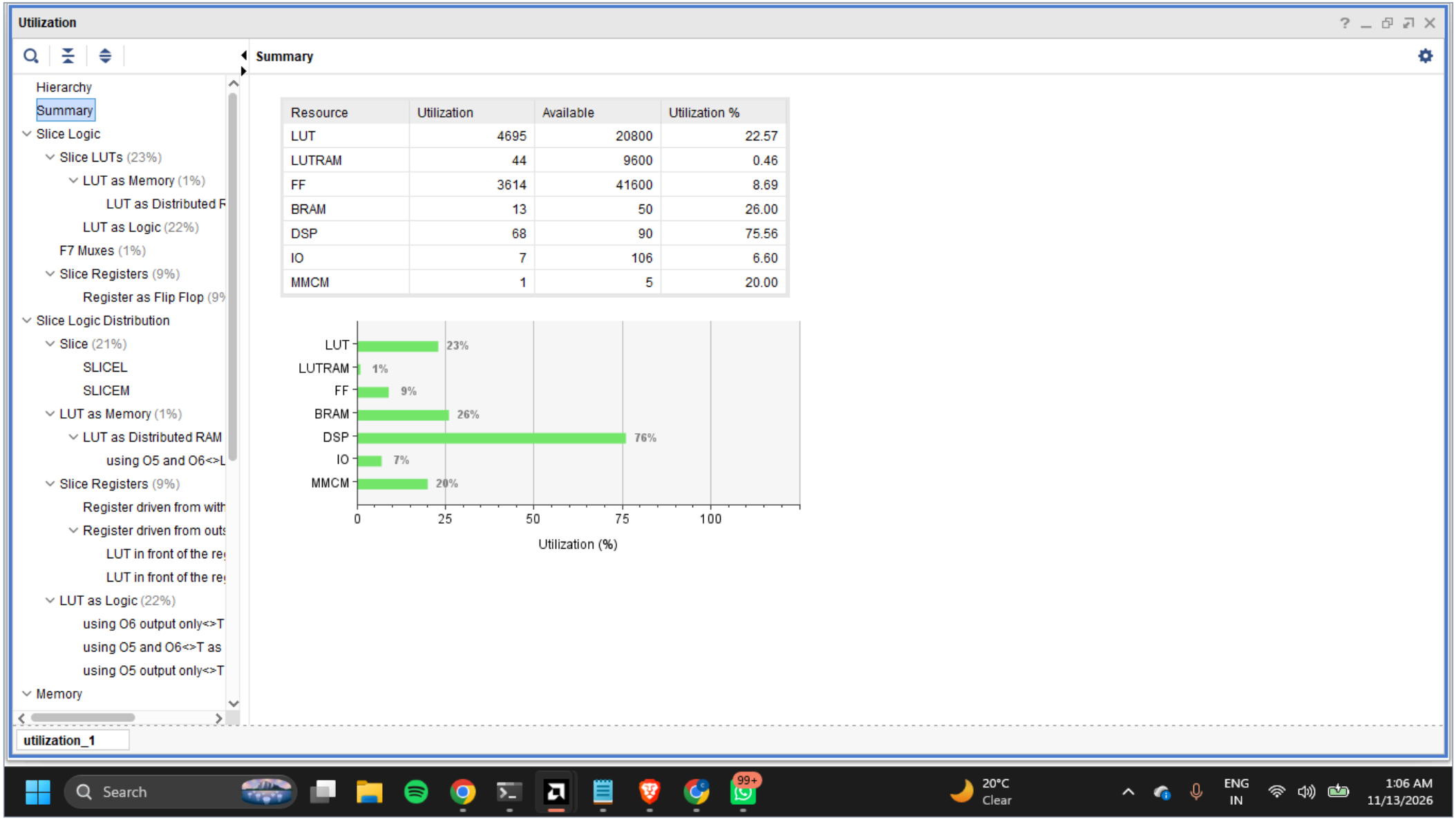
Data Path

Delay Type	Incr (ns)	Path (...	Location	Cell Pin	Cell	Netlist Resources
FDRE (Prop_fdre_C_Q)	(r) 0.456	6.549	Site: SLICE_X49Y28	Q	tap_count_q_reg[1] (FDRE)	u_soc/u_accel_bus/u_accel/u_arch_selector/u_seq_engine/tap_count_q_reg[
net (fo=4, routed)	0.843	7.393				u_soc/u_accel_bus/u_accel/u_arch_selector/u_seq_engine/u_mac/product_o
LUT6 (Prop_lut6_I3_O)	(f) 0.124	7.517	Site: SLICE_X50Y28	O	product_o_i_103 (LUT6)	u_soc/u_accel_bus/u_accel/u_arch_selector/u_seq_engine/u_mac/product_o

Search 20°C Clear ENG IN 1:05 AM 11/13/2026

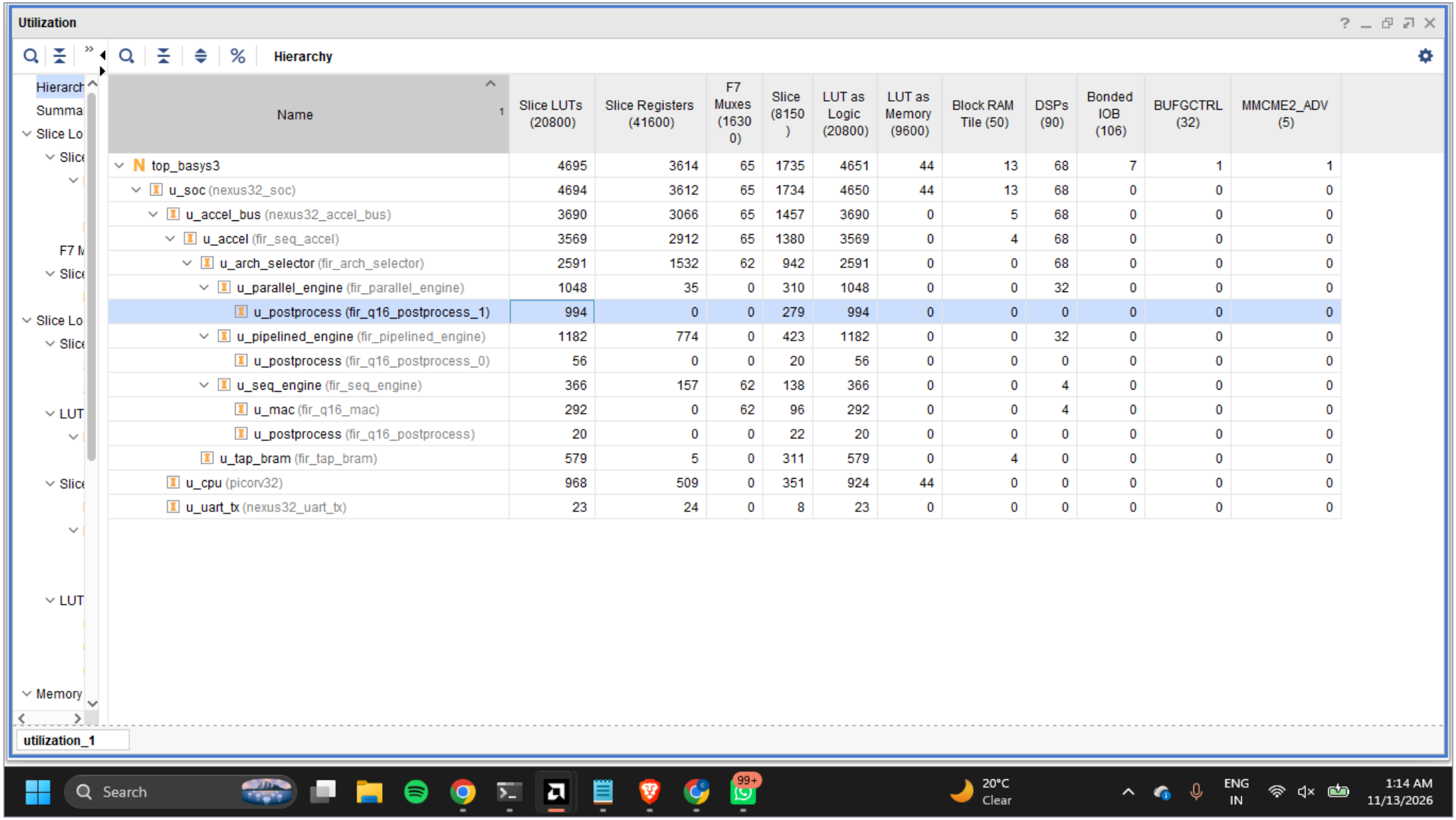
Screenshot 7: Implemented utilization, Summary

LUT 4695 (22.57%), LUTRAM 44, FF 3614 (8.69%), BRAM 13 (26.00%), DSP 68 (75.56%), IO 7, MMCM 1. BRAM, DSP, IO, MMCM identical to synthesis; LUT is 258 lower after optimization and placement. Nothing near 100%.



Screenshot 8: Implemented utilization, Hierarchy (engines expanded)

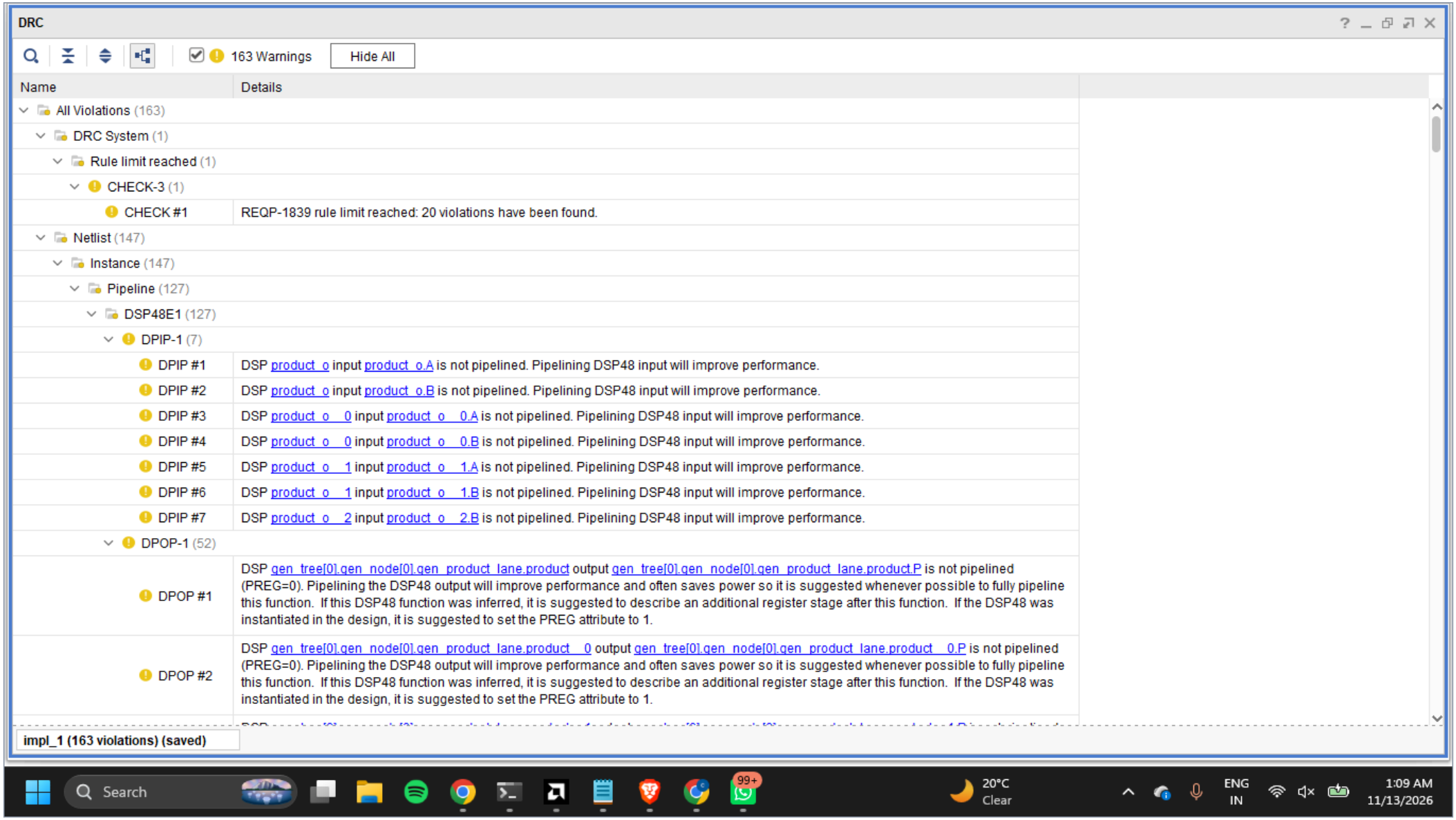
u_arch_selector 2591 LUT / 68 DSP: parallel 1048 LUT / 32 DSP, pipelined 1182 LUT / 32 DSP, sequential 366 LUT / 4 DSP. u_tap_bram 579 LUT / 4 BRAM. u_cpu (picorv32) 968 LUT / 509 FF. u_uart_tx 23 LUT. DSP split 32 + 32 + 4 = 68.



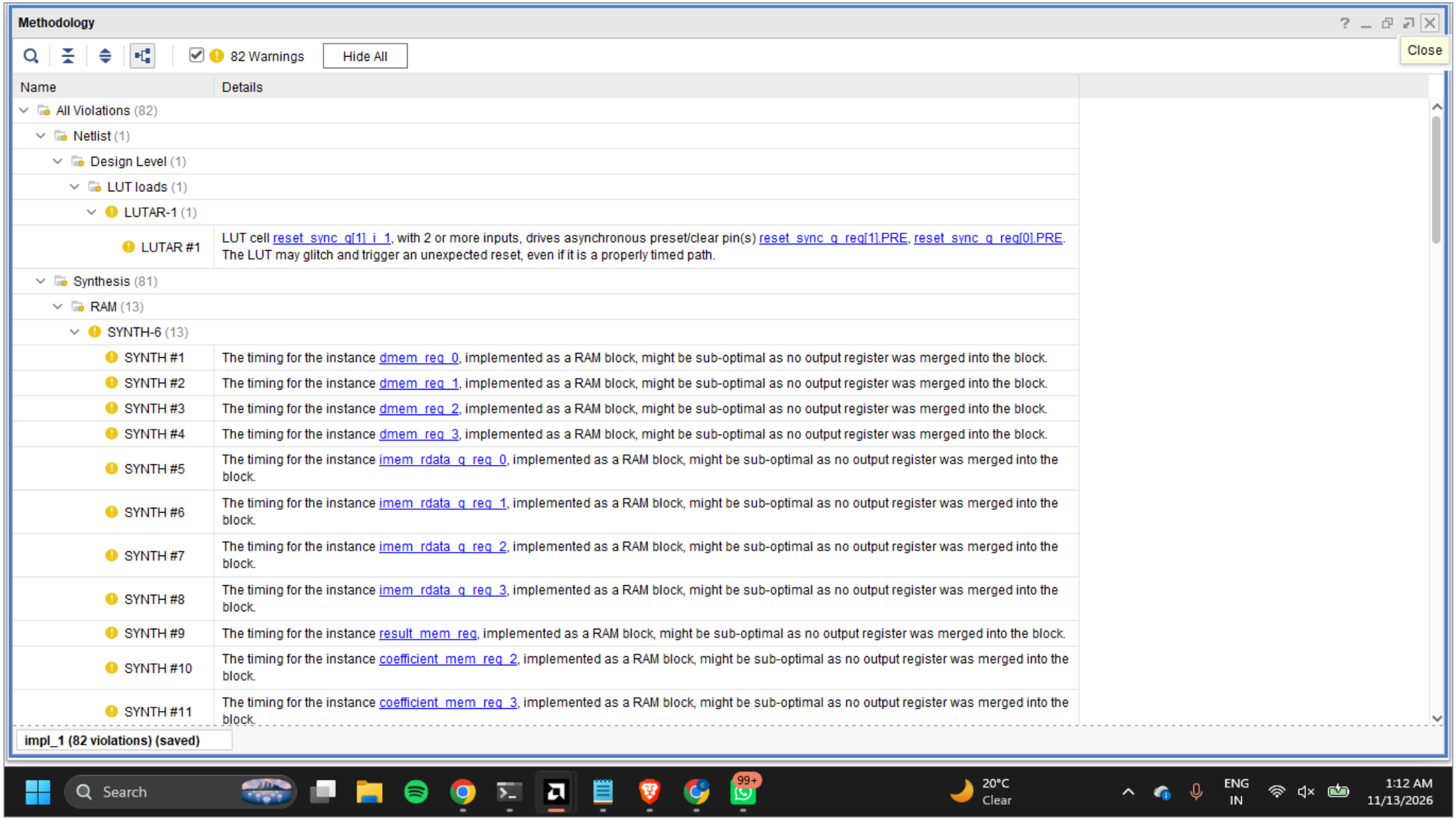
Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (8150)	LUT as Logic (20800)	LUT as Memory (9600)	Block RAM Tile (50)	DSPs (90)	Bonded IOB (106)	BUFGCTRL (32)	MMCME2_ADV (5)
top_basys3	4695	3614	65	1735	4651	44	13	68	7	1	1
u_soc (nexus32_soc)	4694	3612	65	1734	4650	44	13	68	0	0	0
u_accel_bus (nexus32_accel_bus)	3690	3066	65	1457	3690	0	5	68	0	0	0
u_accel (fir_seq_accel)	3569	2912	65	1380	3569	0	4	68	0	0	0
u_arch_selector (fir_arch_selector)	2591	1532	62	942	2591	0	0	68	0	0	0
u_parallel_engine (fir_parallel_engine)	1048	35	0	310	1048	0	0	32	0	0	0
u_postprocess (fir_q16_postprocess_1)	994	0	0	279	994	0	0	0	0	0	0
u_pipelined_engine (fir_pipelined_engine)	1182	774	0	423	1182	0	0	32	0	0	0
u_postprocess (fir_q16_postprocess_0)	56	0	0	20	56	0	0	0	0	0	0
u_seq_engine (fir_seq_engine)	366	157	62	138	366	0	0	4	0	0	0
u_mac (fir_q16_mac)	292	0	62	96	292	0	0	4	0	0	0
u_postprocess (fir_q16_postprocess)	20	0	0	22	20	0	0	0	0	0	0
u_tap_bram (fir_tap_bram)	579	5	0	311	579	0	4	0	0	0	0
u_cpu (picorv32)	968	509	0	351	924	44	0	0	0	0	0
u_uart_tx (nexus32_uart_tx)	23	24	0	8	23	0	0	0	0	0	0

Screenshot DRC: DRC report, routed design

163 warnings, 0 errors, 0 critical. Visible: REQP-1839 rule limit reached (20 violations found), DSP48E1 pipelining advisories DPIP-1 (7 listed) and DPOP-1 (52 listed). Advisory only; the RTL is frozen. REQP-1839 is a known warning. Remaining entries not captured.

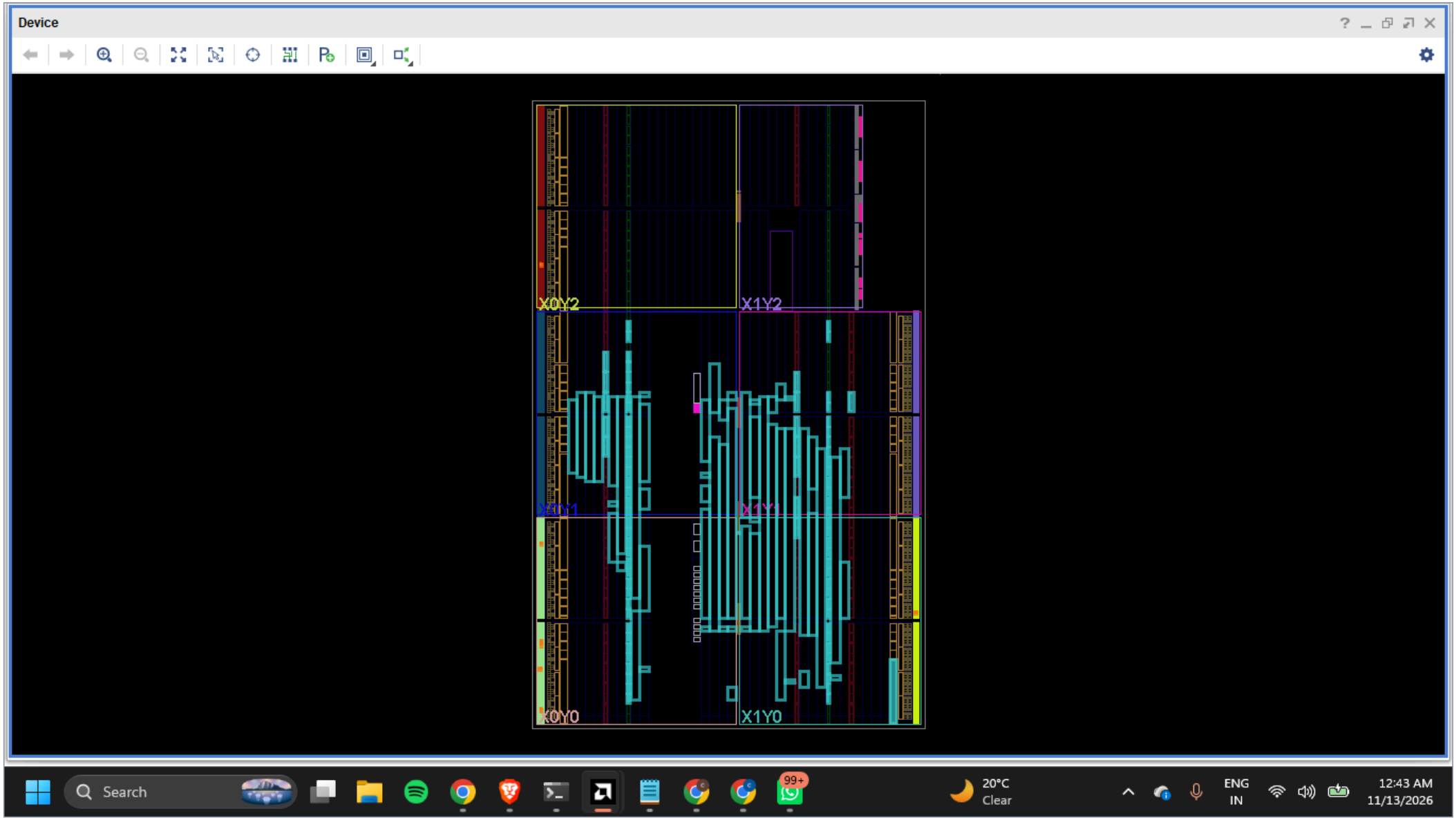


82 warnings, 0 critical. LUTAR-1 (1): LUT in front of the reset synchroniser async pin. SYNTH-6 (13): RAM blocks without a merged output register (dmem, imem, result_mem, coefficient_mem), timing may be sub-optimal. Remaining entries not captured. Advisory only.



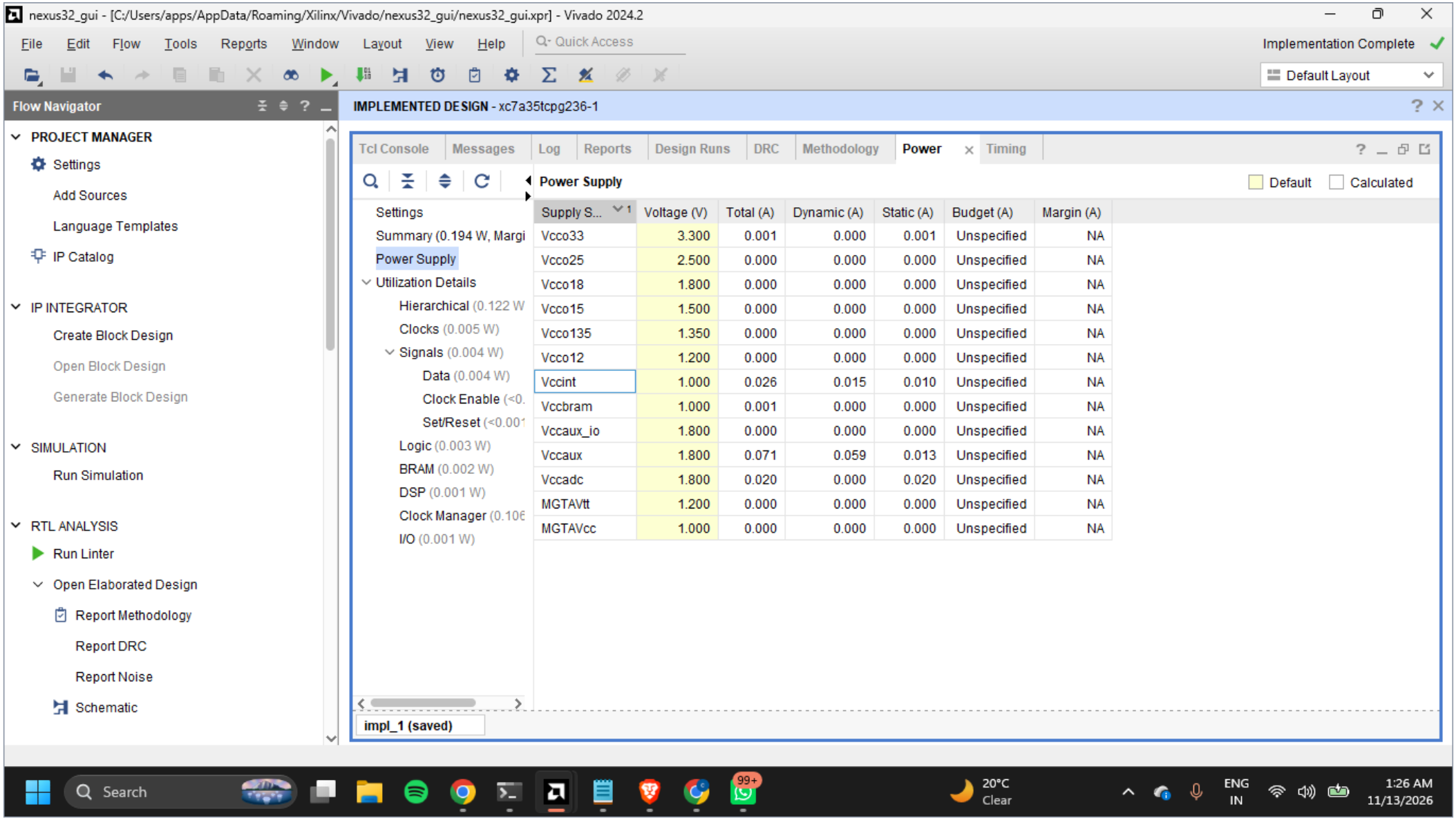
Screenshot 9: Power summary (Vivado estimate, not a measurement)

Total 0.194 W (dynamic 0.122 W, static 0.073 W); junction 26.0 C; thermal margin 59.0 C; confidence Medium. The MMCM is 87% of dynamic power (0.106 W), so power is dominated by clock generation. Vectorless estimate: do not use it to compare the engines.



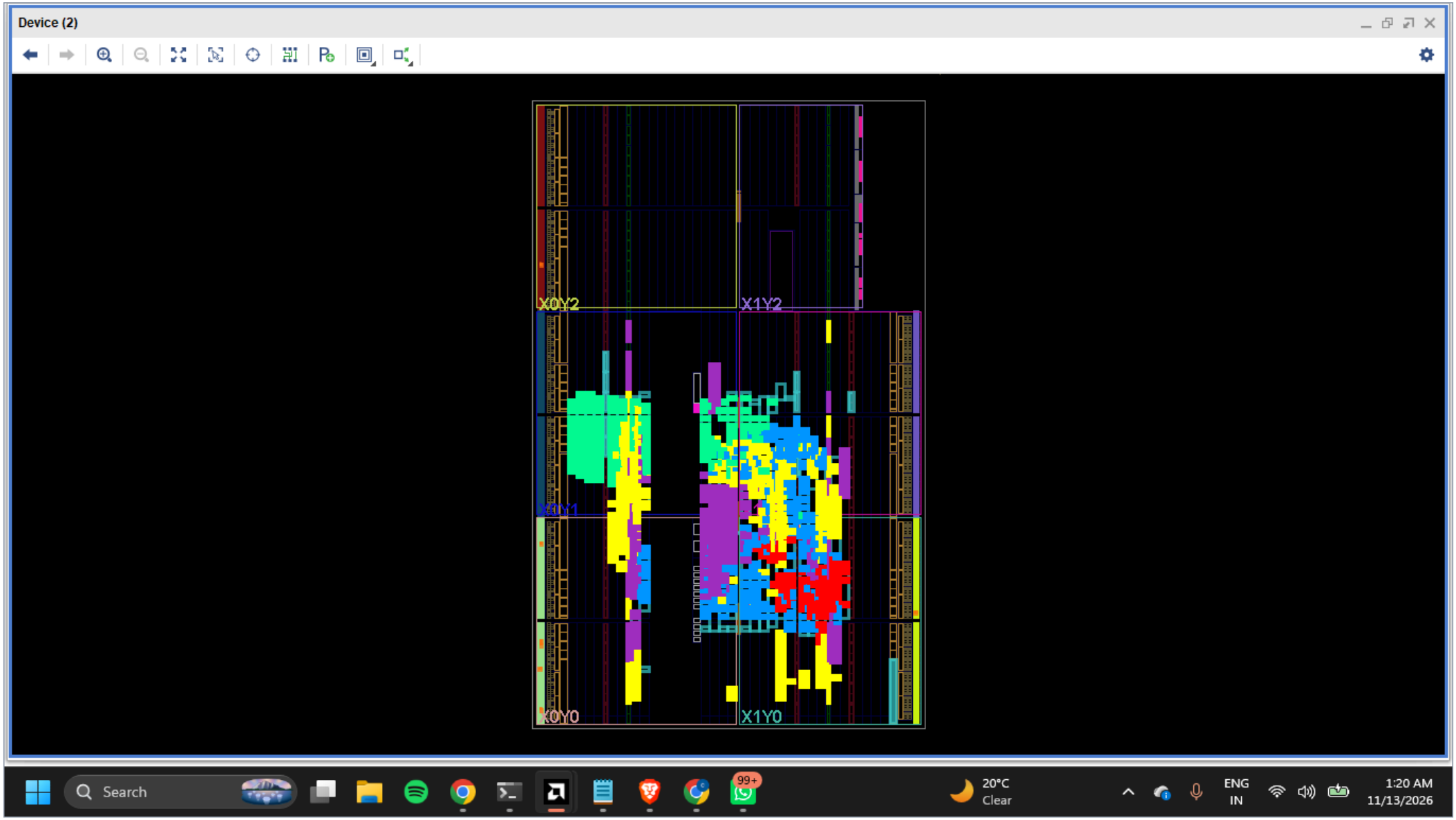
Screenshot 9b: Power supply breakdown (Vivado estimate)

Vccint 1.000 V: 0.026 A total (0.015 dynamic, 0.010 static). Vccaux 1.800 V: 0.071 A (0.059 dynamic; MMCM rail). Vccbram 0.001 A, Vccadc 0.020 A (static), Vcco33 0.001 A. All other rails 0. Estimates, not measurements.



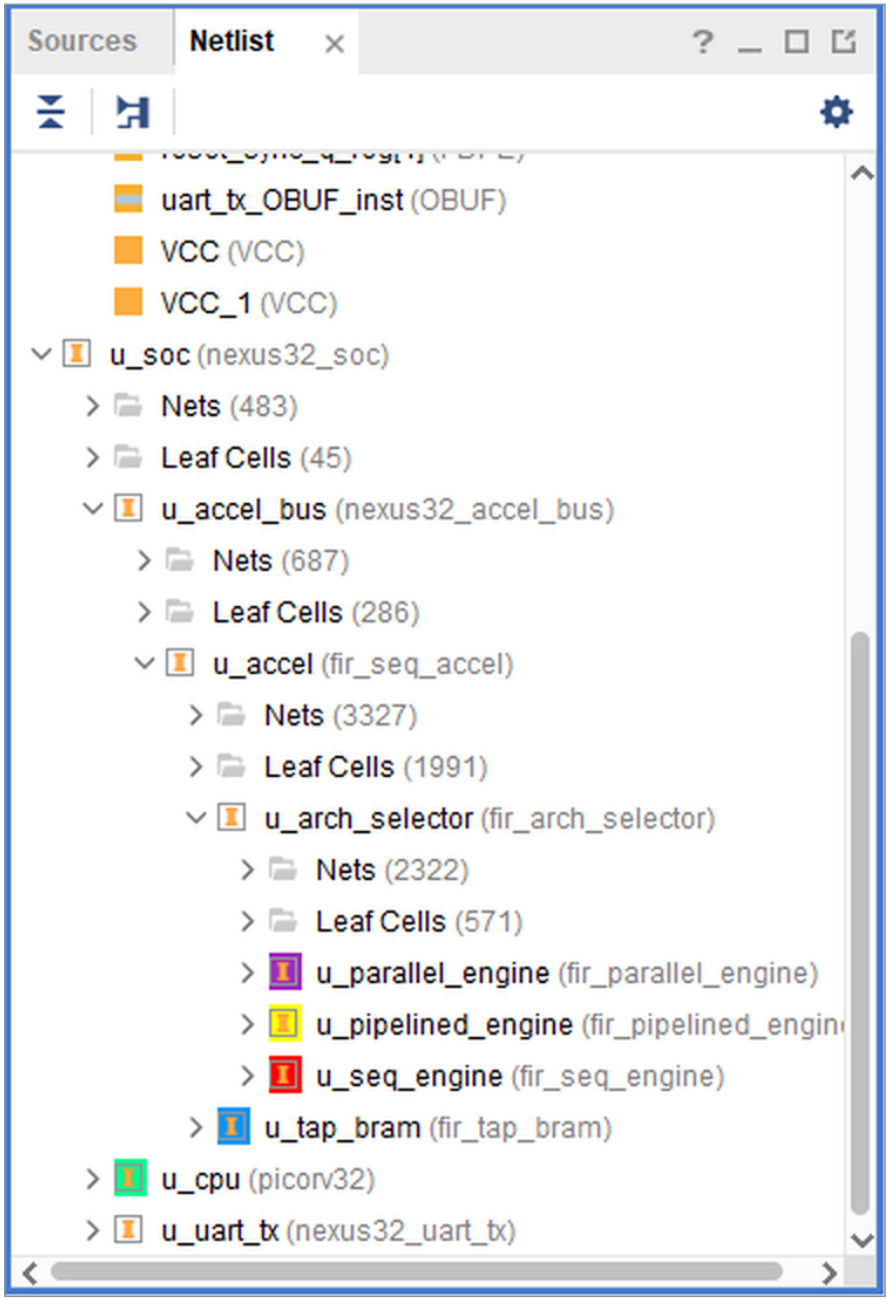
Screenshot 10: Device view, placed and routed, per-engine colours

Red = u_seq_engine (SEQ). Purple = u_parallel_engine (PAR). Yellow = u_pipelined_engine (PIPE). Blue = u_tap_bram. Green = u_cpu (picorv32). Teal outlines = uncoloured cells (UART, bus glue, top level). PAR and PIPE spread into narrow strips because each needs 32 DSP slices from the DSP columns.



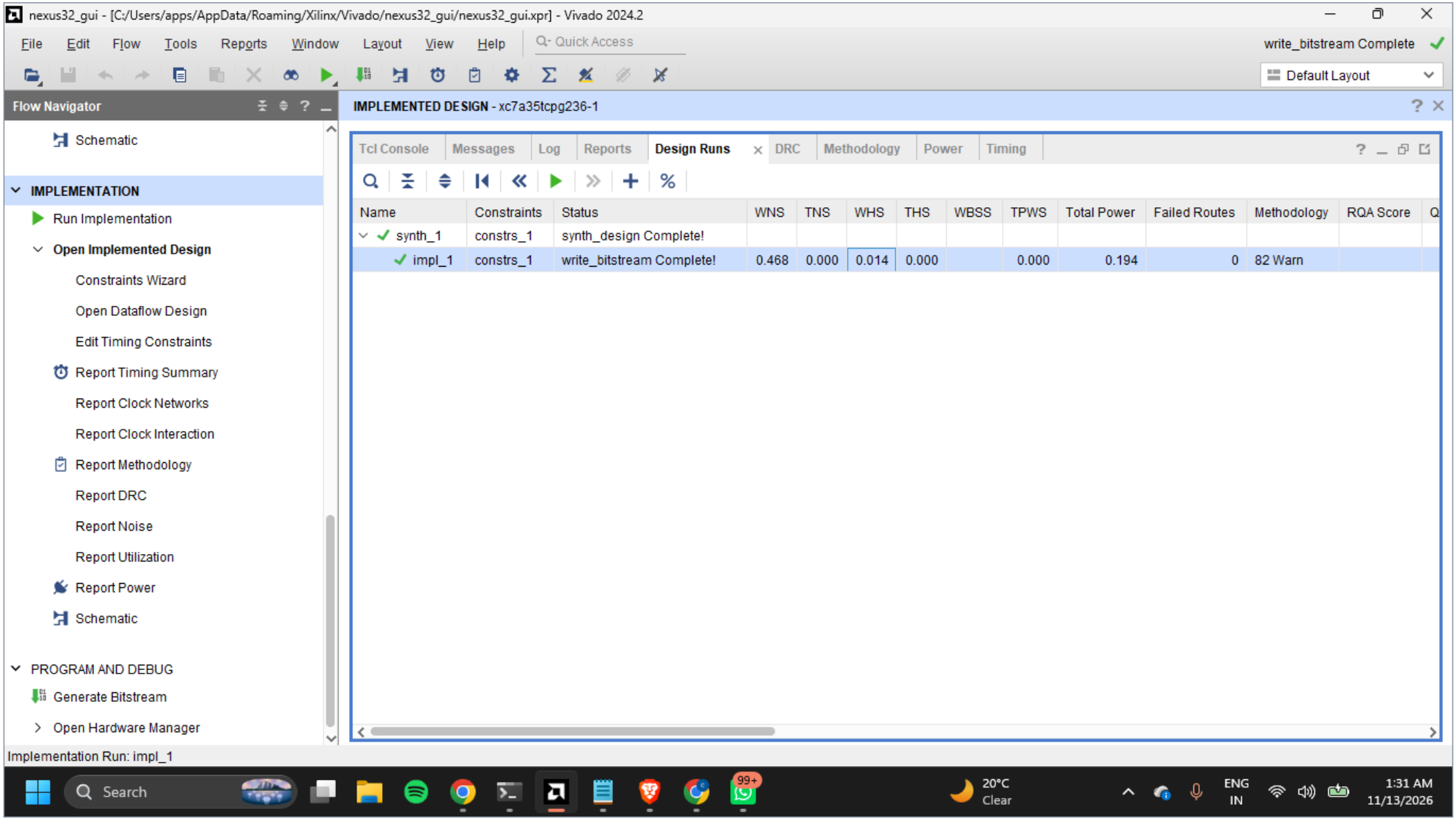
Screenshot 10b: Colour legend for screenshot 10 (Netlist panel)

Cropped from the Netlist window under IMPLEMENTED DESIGN. The coloured icons show the colour assigned to each block: u_parallel_engine purple, u_pipelined_engine yellow, u_seq_engine red, u_tap_bram blue, u_cpu green.



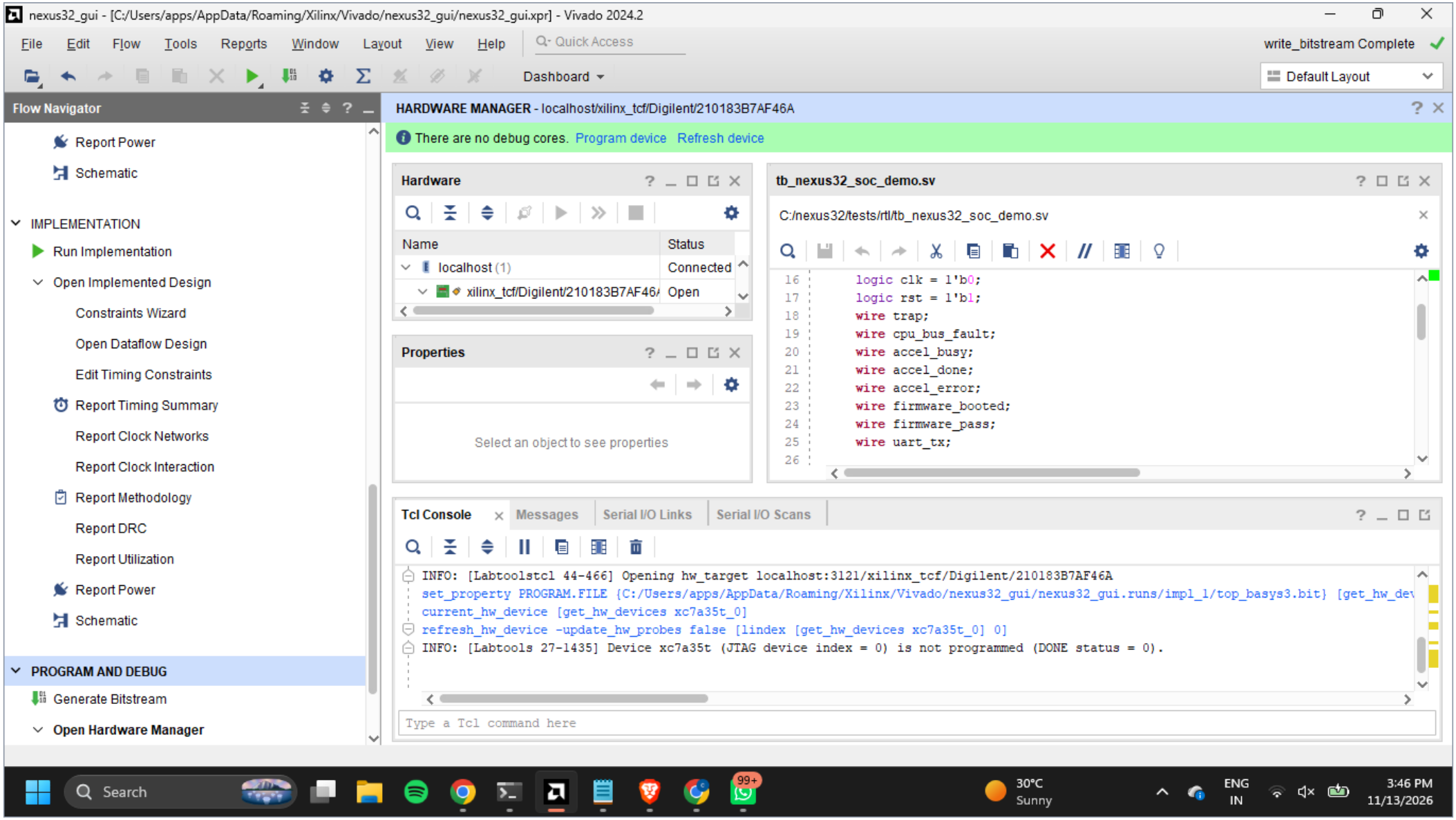
Screenshot BS: Bitstream generated (Design Runs)

impl_1: write_bitstream Complete!, WNS 0.468, TNS 0.000, WHS 0.014, THS 0.000, total power 0.194 W, 0 failed routes, 82 methodology warnings. Generation started 1:28 AM; top_basys3.bit (about 2.08 MB) written 1:29 AM. Timing unchanged by bitstream generation.



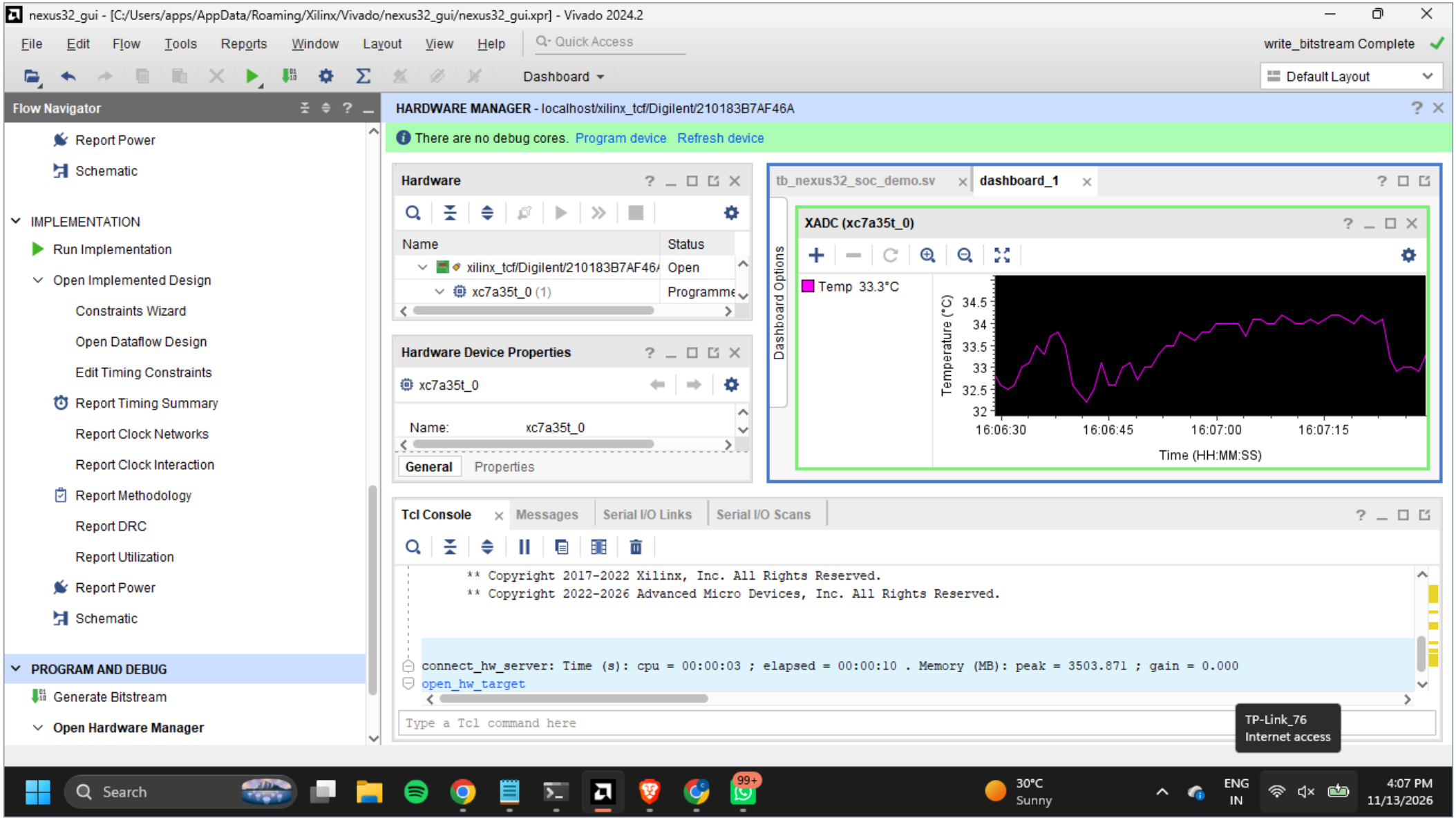
Screenshot HW: Hardware Manager: Basys 3 connected

hw_target xilinx_tcf/Digilent/210183B7AF46A open, xc7a35t_0 detected, "not programmed (DONE status = 0)" before programming. After Program Device the console reported "End of startup status: HIGH" and no supported debug cores (normal, none were added).



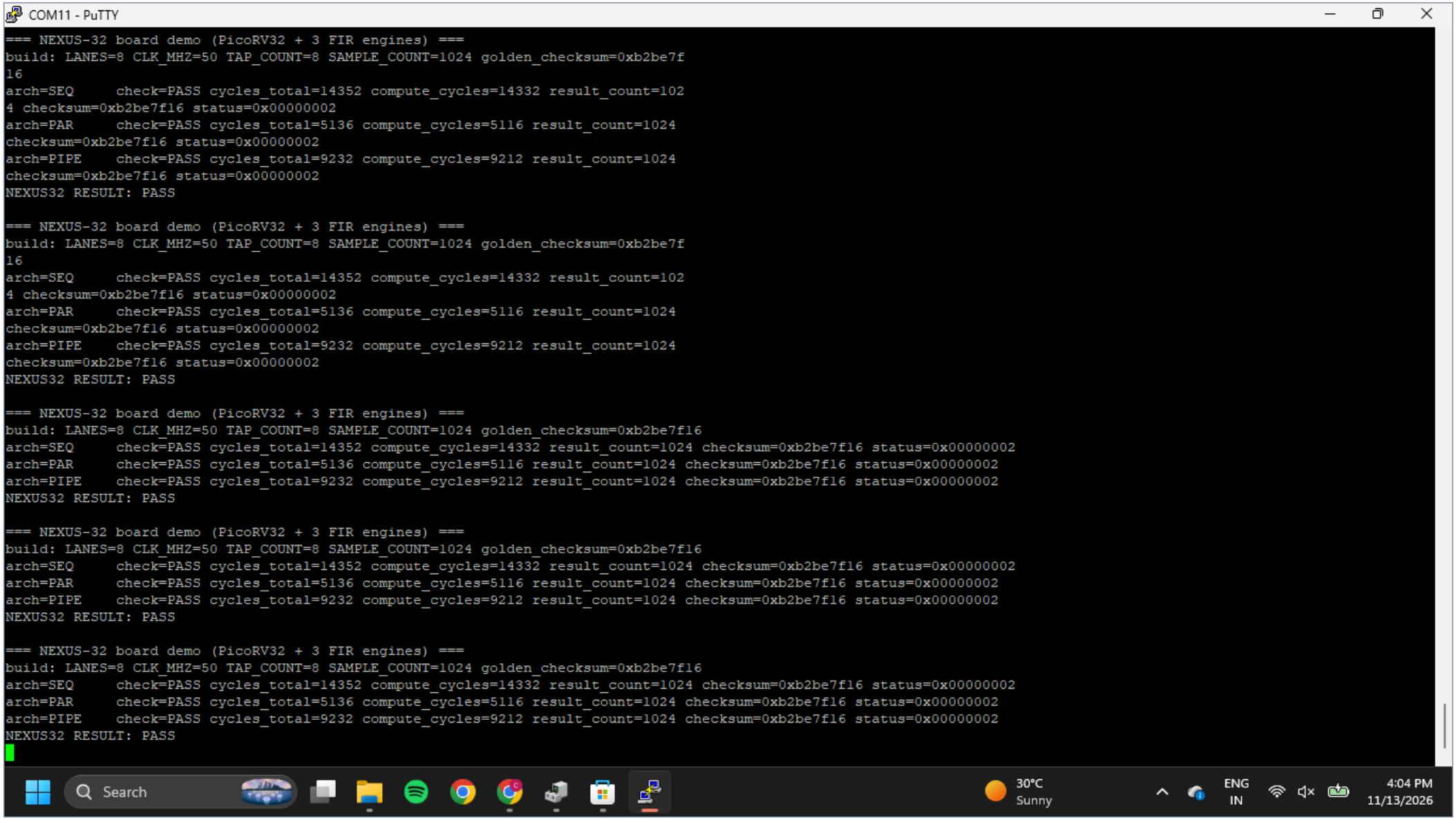
Screenshot 11: System Monitor (XADC) die temperature, measured on the board

Temp 33.3 C at 4:07 PM, chart range about 32 to 34.5 C over roughly one minute, xc7a35t_0 programmed with the LANES=8, 50 MHz bitstream. This is a measurement from the on-die sensor, unlike the 26.0 C Vivado estimate (which assumed 25 C ambient).



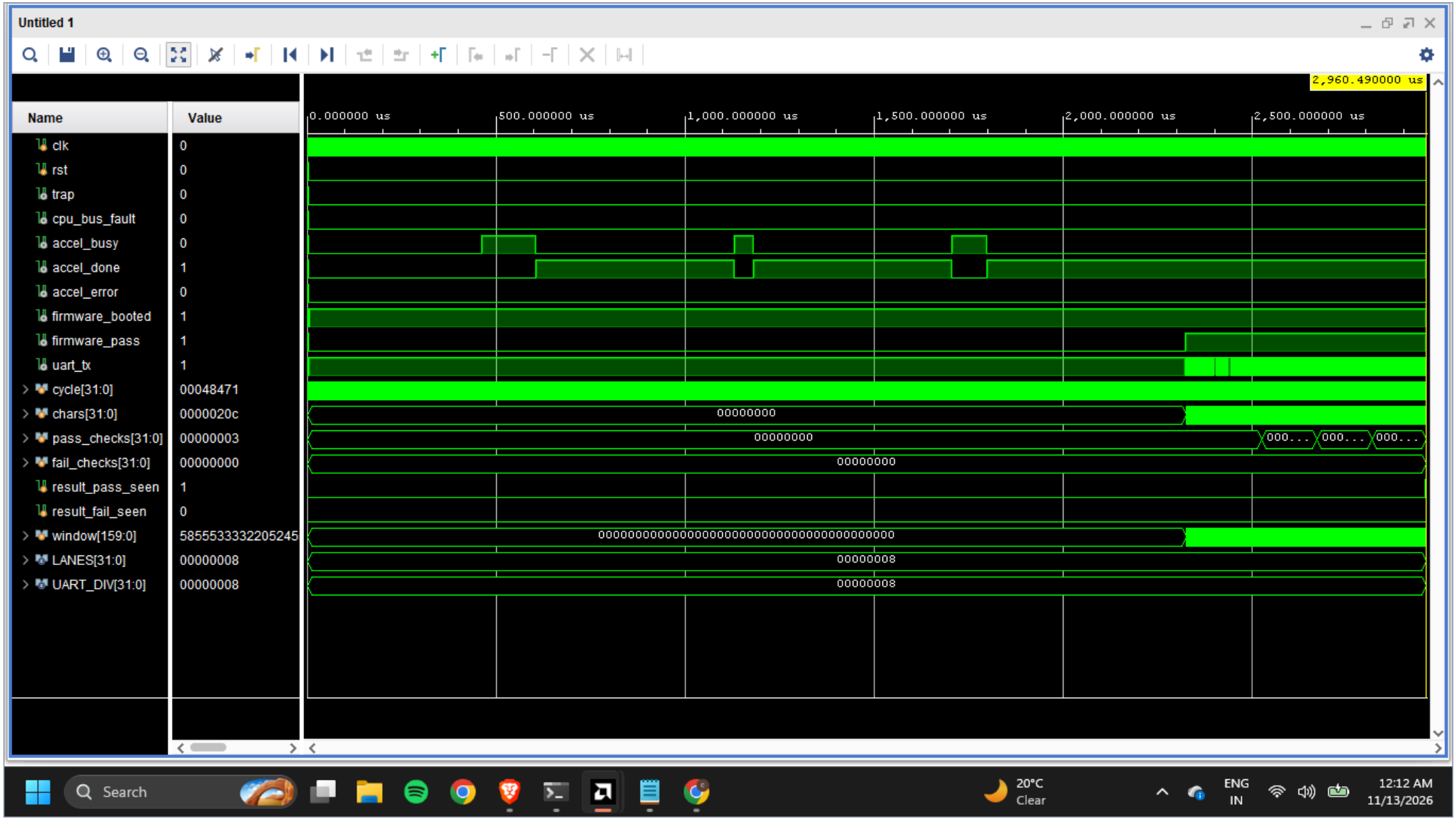
Screenshot 12: UART report from the Basys 3 (PuTTY, COM11, 115200 8N1)

build: LANES=8 CLK_MHZ=50 TAP_COUNT=8 SAMPLE_COUNT=1024. SEQ / PAR / PIPE: check=PASS, cycles_total 14352 / 5136 / 9232, result_count 1024, checksum 0xb2be7f16 (golden), status 0x00000002. NEXUS32 RESULT: PASS. The report repeats (shown five times); the first copies wrap only because the window was narrower then. Identical to the simulation cycle counts.



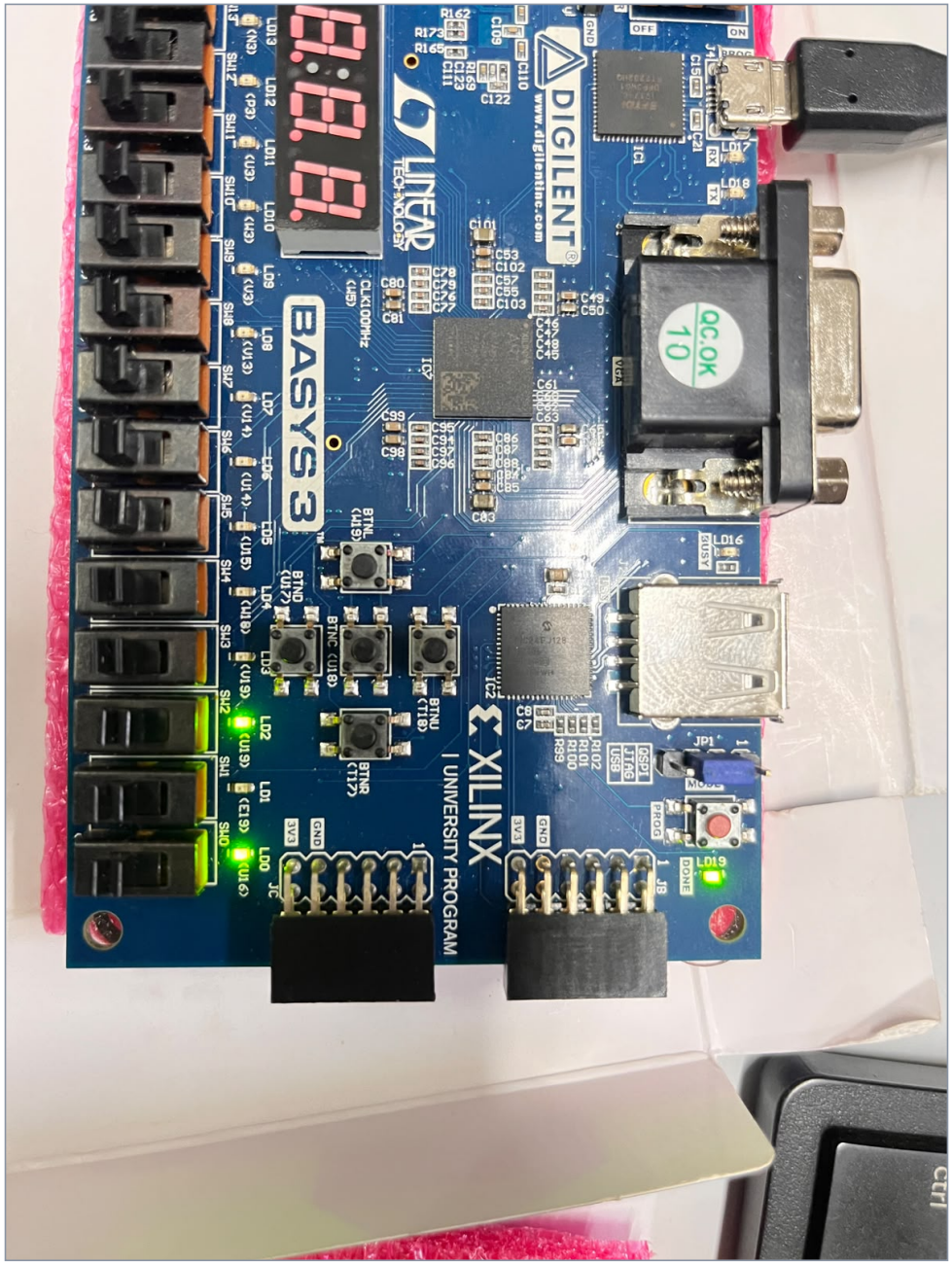
Screenshot S2: Behavioral simulation waveform, LANES=8

Three accel_busy pulses (SEQ longest, PAR shortest). trap, cpu_bus_fault and accel_error stay 0. firmware_pass = 1, pass_checks = 3, fail_checks = 0, 0x20c (524) UART characters. Cycles SEQ 14352 / PAR 5136 / PIPE 9232.



Screenshot Board: Basys 3 programmed and running (LEDs)

After programming top_basys3.bit: DONE (LD19) on, LD0 on, LD1 off, LD2 on, LD3 off, as expected. LD2 alone is not proof of a pass; the UART line (screenshot 12) is. The 7-segment display is not used by this design.



Screenshot Board: Board identification and setup

Silkscreen reads BASYS 3 (Digilent, Xilinx University Program). Micro-USB in the PROG/UART port, red POWER LED on, enumerated as COM11. Photo taken before programming.

